Date	3 July 2026
Team ID	xxxxxx
Project Name	Rising-Waters-Flood-Prediction
Maximum Marks	2 Marks

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR): What the system should do.
- Non-Functional Requirements (NFR): How the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the flood prediction system using authorized login credentials (if login is implemented). If your project has no login, write: "System allows users to access the flood prediction interface securely."	High
2	Authorization levels	Administrator can manage prediction history, while users can perform flood risk prediction. *(If no admin module: "Single user access with prediction functionality.")*	Medium
3	External interfaces	System accepts environmental parameter inputs from users and displays flood prediction results through a web interface.	High
4	Transactions processing		High

		The system processes user inputs, performs ML prediction using the trained XGBoost model, and stores prediction history. The system processes user inputs, performs ML prediction using the trained XGBoost model, and stores prediction history.	
5	Reporting	System displays flood risk percentage and safety recommendations, and stores previous prediction records.	Medium
6	Business rules, etc.	All input values must be valid (e.g., within the expected range) before prediction is performed.	High
7	Compliance to laws or regulations	User data is processed only for prediction purposes and basic privacy practices are followed.	Low
8	Other	Future versions may include real-time weather API integration and SMS/email alerts.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly.	Response time < 3 seconds
2	Scalability	System should support multiple prediction requests.	Support multiple concurrent users
3	Security & Data Privacy	User inputs and prediction history should be stored securely.	No unauthorized access

4	Reliability & Availability	System should provide accurate predictions whenever available.	99% availability during testing
5	Usability & Accessibility	User interface should be simple and easy to use.	Prediction completed in less than 3 clicks
6	Other	System should be maintainable and easy to upgrade.	Modular code structure